UCE809: SEISMIC RESISTANT DESIGN OF STRUCTURES

L	T	P	Cr
3	1	0	3.5

Course Objective: This course will be providing insight into design of structures to withstand earthquake forces and related seismic safety issues.

Earthquake Genesis: Causes of earthquake and propagation, Earthquake occurrence and return period, Characterization of strong ground motions, Seismic hazard assessment, Review of damage in past earthquakes.

Introductory Structural Dynamics: Basic concepts of structural vibrations in Single-Degree and Multi-Degree of Freedom systems

Seismic Analysis of Buildings- Introduction to Indian Standard IS 1893 (Part-1)-2016, Seismic design philosophy, Design response spectrum, Seismic analysis of buildings–Static and Dynamic analysis procedures using codal provisions. Seismic load combinations; Introduction to analysis of masonry-infilled RC buildings.

Seismic Design of Building Components: Concept of ductility for seismic resistance; Capacity based design; Introduction to Indian Standard, IS 13920-2016, Detailing provisions in structural elements using codal provisions.

Course Learning Objectives (CLO)

Upon the completion of this course, the students will be able to:

1. Understand concepts and fundamentals related to seismology & seismic hazard assessment.
2. Perform basic dynamic analysis of single and multi-degree of freedom systems
3. Evaluate lateral forces in buildings for seismic loads.
4. Understand ductile detailing provisions in building components

Text Books:

1. Pankaj Agarwal, andManish Shrikhande, Earthquake Resistant Design of Structures, PHI(2022)
2. Mario Paz and William Leigh, Structural Dynamics(Theory and Computation),Kluwer Academic Publishers, London (2004)
3. Srinivas Vasam and K. Jagannadha Rao, Structural Dynamics and Earthquake Engineering, Publisher: S.K. Kataria & Sons, (2018) ISBN 10: 9350146541 / ISBN 13: 9789350146545

Reference Books:

1. Chopra A.K., Dynamics of structures – Theory and Applications to Earthquake Engineering, Pearson Education (2016).
2. Roberto Villaverde, Fundamental Concepts of Earthquake Engineering, CRC Press, (2009), ISBN 9781420064957

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	30